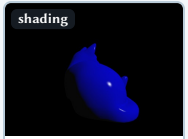
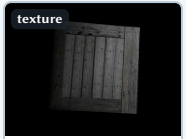
vkmark, 7 scenes

revision ab6e6f3 · 5 causes · repair +135/−58 lines in 18 files

1,124 → 0 timing diagnostics



269 → 0



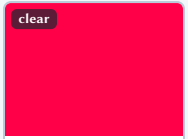
269 → 0



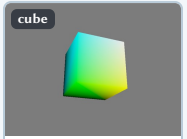
269 → 0



114 → 0



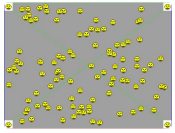
108 → 0



57 → 0



38 → 0



SDL3 sprite test

Vulkan renderer, revision e4af6f8e (Figure 1); a host-write race that validation does not report

testsprite 166

166 → 0 diagnostics

one set of vertex buffers per frame in flight (+38/−18)



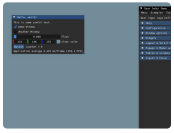
GStreamer, 5 pipelines

gst-plugins-bad 1.24.2, ending in vulkansink

sink	80
colorconvert	118
overlay	80
viewconvert	118
shaderspv	99

495 → 0 diagnostics

two one-line patches



Dear ImGui, 2 examples

Vulkan backend, revision 9d9a621f

glfw	20
sdl2	20

40 → 0 diagnostics

remove one submission (−75)

VKMARK-PSR present semaphore signalled again while a present may wait on it

VKMARK-CLR clear barrier does not chain to the acquire wait

GST-IA sampled image made visible to input-attachment reads

VKMARK-ACQ acquire semaphore reused while its waiting submission may be pending

VKMARK-DEPTH shared depth image ordered for colour only

IMGUI-PRE swapchain images transitioned before they are acquired

VKMARK-UBO host rewrites the uniform buffer a frame in flight reads *(host write)*

GST-WAIT acquire wait at BOTTOM_OF_PIPE blocks nothing

SDL-1 host rewrites vertex buffers frames in flight still read *(host write)*

After the patches, all 15 programs: **no synchronization diagnostic** in either capture mode, no synchronization-validation message, no model warning, and **guarantee-eligible** in the verified configuration; re-tracing each three more times gives the same result.